

Smith Invariants of a Restricted $W(E_6)$ Table-of-Marks Map

Anonymous

Abstract

The preceding finite-group computation gives a nonsingular 16×16 table-of-marks matrix for sixteen core-free subgroup types of $W(E_6)$. We compute its full integral Smith form, refining the determinant to an exact congruence invariant. The Smith invariants are $(1, 2^{\times 10}, 4^{\times 3}, 24, 144)$, so the cokernel is $(\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/4)^3 \oplus \mathbb{Z}/24 \oplus \mathbb{Z}/144$. This is a theorem about the explicitly restricted 16-type mark map; no full Burnside-ring or arithmetic claim is made.

1 Question and scope

Let $G = W(E_6)$ be the order-51840 permutation group fixed by the preceding source package. The C62 atlas and C63 dictionary provide sixteen ambient-conjugacy subgroup types $S_1, \dots, S_{16}$, and C64 defines transitive sets $X_j = G/S_j$ and their self-mark matrix

$$M_{ij} = |(G/S_j)^{S_i}|.$$

C64 proved $\text{rank}_{\mathbb{Q}} M = 16$ and $\det(M) = 226492416 = 2^{23}3^3$. C65 instead studied the rank-three image of the C63 character kernel. Here we ask for the full integral cokernel of M .

The scope is the lattice map generated by these sixteen types only. The firewall is

NO_BAD_EULER_OR_ROOT_NUMBER.

We do not claim a full table of marks, a full Burnside-ring classification, resolvents, local fields, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator.

2 Exact integral calculation

For an integer matrix A , its Smith form is obtained by unimodular row and column operations and has diagonal entries $d_1 \mid d_2 \mid \dots$. The product $d_1 \cdots d_k$ is the k th determinantal divisor. Thus the Smith form records all integral congruence defects, whereas the determinant records only their product.

The C64 matrix is reconstructed from frozen evidence rather than inferred from subgroup orders. Its rows and columns are ordered by the fixed labels $S_1, \dots, S_{16}$. The producer performs exact integer Euclidean row/column reduction. A separate checker repeats the reduction, and a clean SymPy integer Smith implementation supplies a third check.

Theorem 1 (Restricted mark-map Smith form). *For the frozen sixteen C64 subgroup representatives, the self-mark map $m : \mathbb{Z}^{16} \rightarrow \mathbb{Z}^{16}$ has*

$$\text{SNF}(M) = (1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 4, 4, 24, 144).$$

Consequently,

$$\text{coker}(m) \cong (\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/4)^3 \oplus \mathbb{Z}/24 \oplus \mathbb{Z}/144.$$

Certificate. The direct mark reconstruction gives a 16×16 integer matrix of rank 16. Exact Smith reduction returns the displayed diagonal. The successive determinantal divisors are

$$\begin{aligned} &1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 4096, \\ &16384, 65536, 1572864, 226492416. \end{aligned}$$

Their successive ratios are precisely the displayed invariant factors, and the final divisor equals $2^{23}3^3$, the independently recorded C64 determinant. The two independent checkers and library cross-check reproduce all reported invariants. $\square$

Proposition 1 (Primary decomposition). *The 2-primary part of the cokernel is*

$$(\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/4)^3 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/16,$$

and its 3-primary part is $\mathbb{Z}/3 \oplus \mathbb{Z}/9$. No other prime occurs.

Proof. Factor each nontrivial invariant factor in the theorem. The exponents sum to 23 at 2 and 3 at 3, matching the determinant. $\square$

3 Relation to C65 and limitations

C65 certifies the distinct rank-three image lattices

$$\text{SNF}(m\langle z_1, z_3 \rangle) = (2, 8), \quad \text{SNF}(m\langle z_1, z_2, z_3 \rangle) = (2, 2, 8),$$

with relative saturation quotient $\mathbb{Z}/2$. C66 checks these values as an upstream compatibility condition, but does not identify that relative class with a canonical direct summand of $\text{coker}(m)$: such an identification would require an additional explicitly constructed map.

The word “restricted” is essential. The full table of marks of G has more subgroup classes, and this paper neither computes nor classifies that larger object. Likewise, an integral mark cokernel carries no assertion about arithmetic fields or bad-prime factors.

4 Reproducibility

The canonical evidence binds C64 mark evidence, its prefreeze manifest, and C65 saturation evidence by SHA-256. The producer, structural checker, clean-process replay, library SNF cross-check, and ten hostile mutations all pass. The evidence status is `PREFREEZE_G3_PASS`; release closure is separate from the theorem certificate.

5 Conclusion

The C64 determinant $2^{23}3^3$ decomposes into a precise sixteen-factor integral obstruction. This Smith-invariant theorem is the complete integer refinement available from the frozen 16-type mark matrix, while its scope remains deliberately narrower than the full Burnside ring.